

Nadav Halahmi

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Experience

2023 – PRESENT — ISRAELI MINISTRY OF DEFENSE — SOFTWARE ENGINEER

- Research and development of innovative cyber tool as part of an R&D team, across both "home-side" (C#) and "field-side" components (C++).
- Led reliability initiatives: fixed critical core-product bugs and implemented production error monitoring.
- End-to-end development: version compatibility, operational/regular deployments, ongoing support.
- Development in a multi-process environment (gRPC/protobuf), including a plugin architecture.

2020 – 2023 — ISRAELI MINISTRY OF DEFENSE — HPC RESEARCHER

- Accelerated a core software system by ~75% and achieved multicore scalability.
- Improved parallel speedups, applied vectorization, maximized hardware utilization, reduced serial sections, leveraged hyper-threading, and tuned compiler optimizations based on Fortran internals.
- Built a parallel-reproducible runtime configuration while resolving race conditions.
- Led a generic system to automate development processes — received an **efficiency award**.

Education

2016 – 2020 — TECHNION — B.SC. IN SOFTWARE ENGINEERING

- Graduated with distinction (GPA 91); **multiple President's and Dean's honors**.
- **"Final Award"** — awarded annually to 5 top computer science students nationwide.
- **2nd place** — Multi-Agent Search Competition (AI course).

2013 – 2016 — IRONI YUD-DALET HIGH SCHOOL; HEMDA – SCIENCE EDUCATION CENTER

- **Excellence scholarship** in Mathematics and Computer Science (awarded annually to top 3 students).
- **1st place** in the Schwartz-Reisman competition (Computational Science track) - power meter development for cyclists; **2nd place** the following year with a continuation project.

Tools & Technologies

- Languages: C/C++, C#, Python, Fortran, Bash, x86 Assembly, Smali
- Frameworks: gRPC, protobuf, protobuf-net, OpenMP, MPI, Flutter, Unity
- Tools: Intel VTune, Intel Advisor, IDA
- Platforms: POSIX, Windows

Additional Experience & Projects

- 2024 – Present — Invention and development of a strategy board game (physical and digital versions).
- 2024 – Present — Development of an innovative vocabulary-expansion mobile application.
- 2022 – 2023 — Youth Science (Hebrew University) — Course Creator & Lecturer (Cyber).
- 2019 – 2020 — Ironi Yud-Dalet High School — Computer Science Teacher (Grades 10–11).